

Rahul Kumar Kundu

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RESEARCH INTERESTS

- Machine Learning
- Multimodal Learning
- Explainable AI
- Trustworthy AI
- Computer Vision
- Signal Processing
- Medical Image Analysis
- Cybersecurity

RESEARCH SUMMARY

ECE graduate with research experience in machine learning, computer vision, and multimodal learning. Work spans medical image segmentation, fairness and reliability analysis of clinical models, explainable disease classification, and machine learning for cybersecurity. Seeking graduate research opportunities to develop robust, interpretable, and trustworthy AI systems.

ACADEMIC CREDENTIALS

Bachelor of Science (B.Sc.) in Electrical & Computer Engineering (ECE) **January 2020 – July 2025**
Rajshahi University of Engineering & Technology (RUET)
CGPA: **3.79** out of 4.00 | Ranked **7th** out of 60 students | Last 60 Credits CGPA: **3.91**

PUBLICATIONS

Research Profiles: [Google Scholar](#) | [ORCID](#)

Journal Publications

- [1] **R. K. Kundu**, H. B. Kibria, M. F. Ahamed, and M. E. H. Chowdhury, “ED-Net: A Hybrid Deep Learning Framework for Precise Polyp Segmentation Integrating EfficientNetB0 with Double U-Net Architecture and Atrous Spatial Pyramid Pooling with Test-Time Augmentation,” *Biomedical Signal Processing and Control*, vol. 123, Part A, p. 110566, Elsevier, 2026. [**Published, Q1, IF: 5.7**]
DOI: [10.1016/j.bspc.2026.110566](https://doi.org/10.1016/j.bspc.2026.110566)

Conference Proceedings

- [1] R. H. Prince, M. O. F. Goni, H. I. Peyal, and **R.K. Kundu**, “Beyond the Leaf: Intelligent Lightweight Disease Detection in Tea Plants with Spatial Convolutional Neural Networks and Explainable AI,” in *Proc. of the 2025 IEEE 7th International Conference on Sustainable Technologies For Industry 5.0 (STI)*, IEEE, 2025.
DOI: [10.1109/STI69347.2025.11367552](https://doi.org/10.1109/STI69347.2025.11367552)
- [2] S. N. Sakib, O. Jyoti, and **R.K. Kundu**, “Advancing Cybersecurity: A Study on Phishing Website Detection Using Machine Learning,” in *Proc. of the 2025 8th International Conference on Computer and Information Technology (ICCIT)*, IEEE, 2025.
DOI: [10.1109/ICCIT68739.2025.11491653](https://doi.org/10.1109/ICCIT68739.2025.11491653)

Submitted Manuscripts

- [1] H. B. Kibria, **R. K. Kundu**, and N. B. G. Kibria, “Data Leakage in Clinical Machine Learning: A Systematic Review of Sources, Impacts and Mitigation Strategies,” 2026. [**Under Review, Q1**]
- [2] **R. K. Kundu**, M. M. Mia, A. Khandakar, Z. Shi, H. B. Kibria, and M. A. Ayari, “Demographic Fairness in Skin Lesion Segmentation: Lesion-Type Confounding of Age-Related Performance Disparities,” 2026. [**Under Review, Q1**]

Ongoing Research Projects

Q-Cattle: Multimodal Sensor Fusion for Dairy Cattle Behaviour Recognition 2026 – Present

- Developing a multimodal behaviour-recognition framework fusing neck IMU, UWB spatial, ankle-posture, and environmental (THI/CBT) sensor streams, with an IMU-only model intended for practical low-cost deployment.
- Designing leakage-safe leave-one-cow-out evaluation and natural sensor-gap robustness analysis on the MmCows dataset.

RESEARCH EXPERIENCE

Research Assistant **June 2026 – Present**
Supervisor: Dr. Amith Khandakar, Department of Electrical Engineering, Qatar University *Remote*

- Conduct research in machine learning, deep learning, and multimodal learning, contributing to model development, statistical evaluation, reproducible experimentation, and manuscript preparation.

Undergraduate Thesis Research

July 2024 – July 2025

Supervisor: Hafsa Binte Kibria, Assistant Professor, Dept. of ECE, RUET

- Designed **ED-Net** to address joint Dice-IoU optimization, limited cross-dataset generalization, accuracy-efficiency trade-off, and interpretability gaps in existing polyp segmentation models.
- Proposed a hybrid dual-encoder-decoder (EfficientNetB0 + Double U-Net) with ASPP and SE-block attention, achieving **97.13% accuracy**, **92.26% Dice**, and **91.08% IoU** on Kvasir-SEG with TTA (47.17 FPS without TTA, 12.93 FPS with TTA).
- Demonstrated cross-dataset robustness on three external benchmarks (CVC-ClinicDB: 80.62%, ETIS-LaribPolypDB: 77.15%, PolypGen2021: 80.71% Dice) with a dual-mode clinical deployment strategy for real-time screening and offline diagnostic workflows.
- Conducted Grad-CAM and SE block attention analysis confirming clinically interpretable decision-making focused on diagnostically relevant polyp regions.
- Published in *Biomedical Signal Processing and Control*, vol. 123, Part A, p. 110566, 2026, Elsevier (**Q1, IF: 5.7**). DOI: [10.1016/j.bspc.2026.110566](https://doi.org/10.1016/j.bspc.2026.110566)

STANDARDIZED TEST SCORES

Language Proficiency (IELTS)

8th November 2025

Overall Band	Listening	Speaking	Reading	Writing
7.0	7.5	7.0	7.5	6.0

TECHNICAL SKILLS

- Programming Languages:** Python, MATLAB, C, C++, Java.
- ML & DL Frameworks:** TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, pandas, NumPy, Matplotlib.
- Research Tools:** Kaggle, Google Colab, Jupyter Notebooks, Git, GitHub, LaTeX, Mendeley.
- Software:** Visual Studio Code, Simulink, Arduino, Canva, MS Office Suite.
- Web Development:** HTML, CSS, JavaScript, PHP, Bootstrap, MySQL, Next.js, MongoDB.

PROJECTS

- RUET-ASSR Official Website:** Led a team to build the official website for the Astronomy & Science Society of RUET using *Next.js*, *Prisma*, and *MongoDB*. [\[Link\]](#)
- Home Automation System:** Designed an IoT-based home automation system using ESP32 with remote control via Wi-Fi, developed during ECE-3212: Peripherals & Interfacing Sessional. [\[Link\]](#)

ACADEMIC AWARDS AND ACHIEVEMENTS

- Technical Scholarship**, University Grant Commission (UGC) — three consecutive years: 2022, 2023, and 2024, awarded to top-ranked students based on B.Sc. annual results.
- General Grade Scholarship** in SSC (Grade 10) and JSC (Grade 8) examinations, awarded to nationwide highest-ranked students.

ACADEMIC LEADERSHIP & SERVICE

- Chair**, IEEE RUET Signal Processing Society (SPS) Student Branch Chapter (2024–2025) [\[Link\]](#)
- Vice President (Information)**, Astronomy & Science Society of RUET (ASSR) (2024–2025) [\[Link\]](#)
- Secretary, Website & CMT Management**, 2025 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON) (2023–2024) [\[Link\]](#)
- Webmaster**, IEEE RUET Student Branch (2023–2024) [\[Link\]](#)
- Event Management Secretary**, Astronomy & Science Society of RUET (ASSR) (2023–2024) [\[Link\]](#)
- Team Leader**, AURORA National Science Festival 2023 (ASSR) [\[Link\]](#)

REFERENCES

Hafsa Binte Kibria Assistant Professor Department of Electrical and Computer Engineering (ECE), RUET Email: hafsabintekibria@ece.ruet.ac.bd Cell: +8801743289800 Profile: ruet.ac.bd/hafsabintekibria	Md. Nahiduzzaman Assistant Professor Department of Electrical and Computer Engineering (ECE), RUET Email: nahiduzzaman@ece.ruet.ac.bd Cell: +8801763591843 Profile: ruet.ac.bd/mdnahiduzzaman320	Oishi Jyoti Assistant Professor Department of Electrical and Computer Engineering (ECE), RUET Email: oj.ruet@ece.ruet.ac.bd Cell: +8801768787593 Profile: ruet.ac.bd/ojruet
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